MSG No. 150.28 P_c321 [Type IV, trigonal]

Table 1: Wyckoff site: `2a`, site symmetry: `32.`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,5,6]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[7,8,9,10,11,12]`

Table 2: Wyckoff site: `2b`, site symmetry: `32'.`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,10,11,12]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[4,5,6,7,8,9]`

Table 3: Wyckoff site: `4c`, site symmetry: `3..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3]`
2	$[0,\ 0,\ -z]$	`[4,5,6]`
3	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[7,8,9]`
4	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[10,11,12]`

Table 4: Wyckoff site: `4d`, site symmetry: `3..`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,2,3]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[4,5,6]`
3	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{2}\right]$	`[7,8,9]`
4	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}-z\right]$	`[10,11,12]`

Table 5: Wyckoff site: `6e`, site symmetry: `.2.`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,4]`
2	$[0,\ x,\ 0]$	`[2,5]`
3	$[-x,\ -x,\ 0]$	`[3,6]`
4	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[7,10]`
5	$\left[0,\ x,\ \frac{1}{2}\right]$	`[8,11]`
6	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[9,12]`

Table 6: Wyckoff site: `6f`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,10]`

continued ...

Table 6

No.	position	mapping
2	$\left[0,\, x,\, \frac{1}{4}\right]$	[2,11]
3	$\left[-x,\, -x,\, \frac{1}{4}\right]$	[3,12]
4	$\left[x,\, 0,\, \frac{3}{4}\right]$	[4,7]
5	$\left[0,\, x,\, \frac{3}{4}\right]$	[5,8]
6	$\left[-x,\, -x,\, \frac{3}{4}\right]$	[6,9]

Table 7: Wyckoff site: 12g, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[-y,\, x-y,\, z]$	[2]
3	$[-x+y,\, -x,\, z]$	[3]
4	$[x-y,\, -y,\, -z]$	[4]
5	$[y,\, x,\, -z]$	[5]
6	$[-x,\, -x+y,\, -z]$	[6]
7	$\left[x,\, y,\, z+\frac{1}{2}\right]$	[7]
8	$\left[-y,\, x-y,\, z+\frac{1}{2}\right]$	[8]
9	$\left[-x+y,\, -x,\, z+\frac{1}{2}\right]$	[9]
10	$\left[x-y,\, -y,\, \frac{1}{2}-z\right]$	[10]
11	$\left[y,\, x,\, \frac{1}{2}-z\right]$	[11]
12	$\left[-x,\, -x+y,\, \frac{1}{2}-z\right]$	[12]